

Total No. of Questions : 8]

SEAT No. :

P-2902

[Total No. of Pages : 3

[6004]-800

B.E. (Computer Engineering)

MACHINE LEARNING

(2015 Pattern) (Semester - II) (410250)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Discuss the process of [6]

- i) Creating training and test sets
- ii) Managing categorical data

How is scikit-learn library used to manage categorical data?

- b) With the reference to machine learning explain the concept of adaptive machine. What is the difference between classic and adaptive machines? [6]
- c) Explain principal component analysis. Support your answer with suitable example. [8]

OR

Q2) a) Discuss the role of Machine Learning in the following applications: [6]

- i) Virtual Personal Assistants
- ii) Predictions while Commuting

- b) Differentiate with an example label encoder and one hot encoder for managing categorical data. [6]
- c) Explain Logistic regression with suitable example. What is the difference between Logistic regression and Linear regression? [8]

P.T.O.

- Q3)** a) What is *Naive Bayes* classifier? What are the different types of *Naive Bayes* classifier? [4]
- b) Describe support vector machine and its implementation in scikit-learn. [6]
- c) Explain Decision tree classification with suitable example? How it is useful for feature extraction. Elaborate. [8]

OR

- Q4)** a) Explain stochastic gradient descent technique with suitable example. [4]
- b) Write a short note on the following : [6]
- i) Controlled Support Vector Machines
- ii) Support Vector Regression.
- c) What are the different ways for measuring classification performance? Explain AUC-ROC curve in detail. [8]
- Q5)** a) What is the difference between DBSCAN and K-Means algorithm? [8]
- b) Write short note on the following : [8]
- i) Bagging and boosting techniques in Machine Learning.
- ii) AdaBoost and Gradient Tree Boosting.

OR

- Q6)** a) With the reference to clustering, explain the issue of “optimization of clusters”. Explain evaluation method for clustering algorithms. [8]
- b) List Applications, Advantages and Disadvantages of Random Forest. [8]
- Q7)** a) Classify the filter techniques in recommender system. Explain content-based filtering technique with suitable example. [8]
- b) Describe building blocks of a Deep Learning Neural Network. Explain working of Deep Learning. [8]

OR

Q8) a) Write short notes on :

[8]

- i) Content based Systems.
- ii) Naïve User based systems.

b) Explain Hierarchical clustering? Elaborate on advantages of using hierarchical clustering over K-means clustering? **[8]**
